

# Cytokine Cascade Detection via Latent Space Geometry

## Session Summary: Methods, Findings, and Open Challenges

Cytokine-MIL Project

19 May 2026

### Abstract

This report summarizes a working session on detecting cytokine signaling cascades from a MIL-finetuned single-cell encoder. We document three complementary methods investigated: (1) the PBS-RC *directional bias* (**geo**) method for candidate pair discovery, (2) the **cell-type ablation** method for directionality determination — found to be the strongest tool of the three — and (3) an alignment-based **inner-product** / **cosine** method for symmetric pair detection. Diagnostic experiments on 8 random seeds showed that *pair detection itself is the current bottleneck*: of 11 pre-registered known cascades, only 4 are recovered in any direction by the geo method at the 20% top threshold, and 0 in the forward direction at the 5% threshold under per-source top-3 filtering. The ablation method works reliably when handed a viable pair list but cannot solve the upstream pair-selection problem. The session ends with the implementation of an alignment-based detection variant designed to address this bottleneck.

## 1 Session Goal and Context

The cytokine-MIL project trains an attention-based multiple-instance-learning classifier on pseudotubes of PBMCs treated with one of 91 cytokines. The hypothesis is that the trained model’s latent geometry and dynamics encode information about cytokine signaling *cascades* — cases where cytokine  $A$  induces a cellular state that subsequently produces cytokine  $B$ , with some intermediate cell type  $T$  acting as a *relay*.

A cascade triple is therefore  $(A, B, T)$ : the source cytokine, the target cytokine, and the relay cell type. The session focused on a two-stage cascade detection pipeline already implemented (geo  $\rightarrow$  ablation), diagnosed multiple failure modes, applied iterative fixes, and finally designed an alternative pair-detection method.

**Validation ground truth.** Eleven canonical cascade pairs (KNOWN\_CASCADES in `scripts/analyze_two_stage_res`) serve as the held-out validation set:

IL-12  $\rightarrow$  IFN- $\gamma$ , IL-1 $\beta$   $\rightarrow$  IL-6, IL-2  $\rightarrow$  IL-15, IL-33  $\rightarrow$  IL-13, IL-18  $\rightarrow$  IFN- $\gamma$ , IL-21  $\rightarrow$  IL-10,  
TNF- $\alpha$   $\rightarrow$  IL-6, IFN- $\alpha$ 1  $\rightarrow$  IFN- $\gamma$ , IL-10  $\rightarrow$  IL-6, IL-4  $\rightarrow$  IL-13, IL-27  $\rightarrow$  IFN- $\gamma$ .

**Experiment grid.** All experiments use 8 random seeds (42, 123, 7, 456, 789, 314, 271, 618) with donors D1, D4–D12 for training and D2, D3 held out for validation. Each experiment trains Stage 1 (encoder pre-training), Stage 2 (MIL with frozen encoder), and Stage 3 (MIL with unfrozen encoder).

## 2 Methods Investigated

### 2.1 Method 1: PBS-RC Directional Bias (“geo”)

**Goal.** Identify ordered candidate pairs  $(A \rightarrow B)$  from the latent space, with a candidate relay cell type  $T$ .

**Definition.** For each tube of cytokine  $C$ , donor  $d$ , and cell type  $T$ , the encoder produces a per-cell-type mean embedding  $\mu_{C,T,d} \in \mathbb{R}^{128}$ . After subtracting the donor-pooled PBS centroid for  $T$  (PBS-RC), the directional bias of  $A \rightarrow B$  via  $T$  for donor  $d$  is:

$$b_{\text{fwd}}^{(d)}(A \rightarrow B, T) = (\mu_{A,T,d} - \mu_A) \cdot \hat{u}_{A \rightarrow B}, \quad \hat{u}_{A \rightarrow B} = \frac{\mu_B - \mu_A}{\|\mu_B - \mu_A\|}$$

where  $\mu_A$  is the pooled PBS-RC centroid of  $A$  across cell types (removing  $A$ 's generic effect, leaving only the  $T$ -specific cascade component). The forward call uses a one-sided Wilcoxon signed-rank test across donors with Bonferroni correction over cell types per ordered pair.

**Encoder choice.** Initial diagnostic compared two encoders on identical inputs:

- **Stage 1** encoder — trained only on cell-type classification. Contains zero cytokine-specific information by construction.
- **Stage 3** encoder — MIL-finetuned. Cells are embedded in a cytokine-discriminative space.

## 2.2 Method 2: Cell-Type Ablation (the directional workhorse)

**Goal.** Given a candidate pair  $(A, B)$ , determine the direction ( $A \rightarrow B$ ,  $B \rightarrow A$ , or shared) and identify the relay cell type  $T^*$ .

**Definition.** For a tube treated with cytokine  $A$  (target  $B$ ), compute the bag-level softmax probability  $P(B \mid \text{tube}_A)$  at the full tube and at the tube with cell type  $T$  ablated (all cells of type  $T$  removed). The relay score is:

$$\text{relay}(A \rightarrow B, T) = P(B \mid \text{tube}_A^{\text{full}}) - P(B \mid \text{tube}_A^{\setminus T})$$

A high positive value means cells of type  $T$  in an  $A$ -treated tube are *necessary* for the model to predict  $B$  — direct functional evidence of  $A \rightarrow B$  via  $T$ .

The direction call compares max-relay across cell types in both directions ( $A \rightarrow B$  vs  $B \rightarrow A$ ). The pipeline does this for both forward and reverse direction per  $(A, B)$  pair, so the ablation truly tests directionality rather than assuming it from geo.

### Why ablation is the strongest method we have.

The ablation method has the property that **the model itself decides** whether the cells of type  $T$  in an  $A$ -treated tube carry the signal for  $B$ . It is a direct functional test of the cascade hypothesis: “does removing  $T$  cells from  $A$ 's tube destroy the model's belief in  $B$ ?” This sidesteps all assumptions about how the latent space encodes cascades. As long as the model successfully learns to discriminate cytokines, ablation gives a faithful direction signal. The cost is computational: per pair  $(A, B)$ , each direction, each cell type  $T$ , each donor, requires one forward pass —  $\sim 3.5$  minutes per ordered pair on the cluster (CPU,  $n=10$  donors  $\times$   $n=18$  cell types  $\times$   $n=10$  tubes per cytokine).

## 2.3 Method 3: Alignment-Based Inner-Product / Cosine (new, designed this session)

**Goal.** A pair-detection method that does *not* bake in a directional assumption (symmetric scoring), addressing the pair-selection bottleneck identified in this session.

**Definition.** For each cytokine  $C$ , cell type  $T$ , donor  $d$ , compute the PBS-RC centroid  $v_{C,T,d} = \mu_{C,T,d} - \mu_{\text{PBS},T}$ . For each unordered pair  $\{A, B\}$ , each  $T$ , each  $d$ :

$$\cos_d(A, B, T) = \frac{\langle v_{A,T,d}, v_{B,T,d} \rangle}{\|v_{A,T,d}\| \cdot \|v_{B,T,d}\|} \quad (\text{scale-free angular alignment})$$

$$\text{ip}_d(A, B, T) = \langle v_{A,T,d}, v_{B,T,d} \rangle \quad (\text{magnitude-weighted})$$

Aggregate across donors by mean, then take max over  $T$  for the pair-level score. The relay  $T^*(A, B)$  is the argmax cell type.

**PCA-denoised variants.** The hypothesis is that of the 128 embedding dimensions, only a small fraction carry cascade-relevant signal. To test this, we project the centroids to lower dimensions (6 and 24) via global PCA on the stacked  $\{v_{C,T,d}\}$  matrix, and score in each subspace.

This produces six variants:  $\{\text{cosine, inner product}\} \times \{6\text{D}, 24\text{D}, 128\text{D}\}$ .

### 3 Diagnostic Results

#### 3.1 Encoder choice: Stage 1 vs Stage 3 for geo detection

The MIL-finetuned (Stage 3) encoder is substantially more informative than the cell-type-only (Stage 1) encoder for cascade detection (Table 1).

Table 1: Stage 1 vs Stage 3 encoder geo recall on KNOWN\_CASCADES (exp\_0\_seed42, all metrics on top-K%). Stage 3 doubles recall at every threshold under almost every metric.

| Metric     | Stage | Recall@5%   | Recall@10%  | Recall@20%  |
|------------|-------|-------------|-------------|-------------|
| max-W      | S1    | 5/11        | 5/11        | 5/11        |
| max-W      | S3    | 4/11        | 6/11        | <b>8/11</b> |
| median-W   | S1    | 0/11        | 1/11        | 2/11        |
| median-W   | S3    | 2/11        | 4/11        | <b>8/11</b> |
| mean-W     | S1    | 1/11        | 1/11        | 4/11        |
| mean-W     | S3    | <b>4/11</b> | <b>7/11</b> | 7/11        |
| n_ct_at_55 | S1    | 3/11        | 5/11        | 5/11        |
| n_ct_at_55 | S3    | 3/11        | 6/11        | 7/11        |

#### 3.2 The W-saturation problem

The original geo ranking used the **maximum** Wilcoxon  $W$  statistic across cell types per pair. With  $n = 10$  donors,  $W_{\max} = 55$ , and with 18 cell types, this maximum is reached for  $\sim 14\%$  of pairs — the top of the ranking is saturated and provides no discrimination between pairs.

##### Quantifying W-saturation.

At Stage 3, of 8010 non-PBS ordered pairs:

- $W_{\max} = 55$  for **13.7%** of pairs (1100 pairs all tied)
- $W_{\max} \geq 50$  for 32.5% of pairs
- median- $W$  ranges 5–50, with much better spread (top 100 has 25 unique values)
- mean- $W$  ranges 23–38, top 100 has 44 unique values

**Fix:** Switched ranking metric from max- $W$  to **mean- $W$  across cell types**. Mean- $W$  has the highest pair discrimination and the best recall@5% (4/11 known cascades).

### 3.3 Per-source filtering: too aggressive at $K = 3$

To diversify the candidate set and ensure every cytokine gets representation in the ablation budget, we tried **per-source top-K** filtering: for each source cytokine  $A$ , keep its top- $K$  targets  $B$  by score. With  $K = 3$  and 90 cytokines, this yields  $\sim 270$  ordered pairs per experiment (manageable for ablation).

**Failure mode: known cascade pairs rank too low per source.**

For most known cascades, the cascade pair does not rank in the source cytokine’s top-3 (Table 2). Stronger “geometric outliers” (e.g., IL-32- $\beta$  with its distinctive B-cell-memory signature) dominate every source’s top-3, even though they are unlikely to be biological cascade partners. **0 of 11 known cascades appear in the per-source top-3 forward direction list; only 2 of 11 appear in any direction.**

Table 2: Source cytokine’s top-3 target list (exp\_0\_seed42, Stage 3, mean-W ranking). For 9 of 11 known cascades, neither the source’s top-3 nor the target’s top-3 includes the cascade partner.

| Known                                       | Source’s top-3 targets                  | Target’s top-3 sources                       | Found?  |
|---------------------------------------------|-----------------------------------------|----------------------------------------------|---------|
| IL-12 $\rightarrow$ IFN- $\gamma$           | C5a, CT-1, TL1A                         | IL-33, IL-8, IFN- $\alpha$ 1                 | no      |
| IL-1 $\beta$ $\rightarrow$ IL-6             | IFN- $\lambda$ 2, Leptin, FGF- $\beta$  | TNF- $\alpha$ , IL-9, PRL                    | no      |
| IL-2 $\rightarrow$ IL-15                    | IFN- $\gamma$ , IL-4, IFN- $\beta$      | IFN- $\beta$ , IFN- $\gamma$ , IFN- $\omega$ | no      |
| IL-33 $\rightarrow$ IL-13                   | CD27L, IL-16, IFN- $\epsilon$           | IL-20, IL-27, IL-1 $\alpha$                  | no      |
| IL-18 $\rightarrow$ IFN- $\gamma$           | GITRL, CT-1, IFN- $\lambda$ 2           | IL-33, IL-8, IFN- $\alpha$ 1                 | no      |
| IL-21 $\rightarrow$ IL-10                   | IL-20, IL-22, ADSF                      | TGF- $\beta$ 1, IFN- $\lambda$ 1, CD30L      | no      |
| TNF- $\alpha$ $\rightarrow$ IL-6            | C3a, IL-31, IL-24                       | <b>TNF-<math>\alpha</math></b> , IL-9, PRL   | reverse |
| IFN- $\alpha$ 1 $\rightarrow$ IFN- $\gamma$ | M-CSF, RANKL, IL-17A                    | IL-33, IL-8, <b>IFN-<math>\alpha</math>1</b> | reverse |
| IL-10 $\rightarrow$ IL-6                    | TGF- $\beta$ 1, IFN- $\lambda$ 1, CD30L | TNF- $\alpha$ , IL-9, PRL                    | no      |
| IL-4 $\rightarrow$ IL-13                    | IL-34, FLT3L, IFN- $\beta$              | IL-20, IL-27, IL-1 $\alpha$                  | no      |
| IL-27 $\rightarrow$ IFN- $\gamma$           | IL-3, FGF- $\beta$ , IL-36Ra            | IL-33, IL-8, IFN- $\alpha$ 1                 | no      |

### 3.4 Ablation results on the per-source top-3 candidates

Despite the geo selection missing most known cascades, the ablation step on the chosen pairs identified several reproducible novel pairs (Table 3).

Table 3: Pairs detected in  $\geq 5$  of 8 experiments via mean-W per-source top-3 with ablation directionality. Numbers show experiment-level vote counts ( $A \rightarrow B : B \rightarrow A$ ).

| Pair                                  | n exps | $A \rightarrow B : B \rightarrow A$ | Biological interpretation                       |
|---------------------------------------|--------|-------------------------------------|-------------------------------------------------|
| IL-9 $\rightarrow$ IL-10              | 6      | 5:1                                 | Treg / mast cell circuit                        |
| IL-23 $\rightarrow$ IFN- $\lambda$ 1  | 6      | 3:3                                 | Th17 $\rightarrow$ antiviral interferon         |
| Noggin $\rightarrow$ IL-10            | 5      | 1:4                                 | BMP antagonist $\rightarrow$ anti-inflammatory  |
| Decorin $\rightarrow$ TRAIL           | 5      | 3:2                                 | ECM proteoglycan $\rightarrow$ apoptosis        |
| IL-15 $\rightarrow$ IL-1 $\beta$      | 5      | 3:2                                 | NK activation $\rightarrow$ inflammasome        |
| IL-36Ra $\rightarrow$ FasL            | 5      | 4:1                                 | IL-36 antagonist $\rightarrow$ apoptosis        |
| IL-36Ra $\rightarrow$ IL-19           | 5      | 4:1                                 | IL-36 antagonist $\rightarrow$ IL-20 subfamily  |
| IL-1 $\alpha$ $\rightarrow$ CT-1      | 5      | 4:1                                 | Alarmin $\rightarrow$ cardiotrophin-1           |
| IL-1 $\alpha$ $\rightarrow$ GM-CSF    | 5      | 1:5                                 | Alarmin / myeloid expansion                     |
| GM-CSF $\rightarrow$ IFN- $\lambda$ 1 | 5      | 1:4                                 | Myeloid activation $\rightarrow$ IFN- $\lambda$ |
| GM-CSF $\rightarrow$ IL-27            | 5      | 2:3                                 | Myeloid activation $\rightarrow$ IL-27          |

**The strongest novel discovery: IL-9  $\rightarrow$  IL-10.**

Detected in 6 of 8 experiments, with 5 of 6 informative votes calling direction  $A \rightarrow B$ . IL-9 is a known driver of Treg expansion and IL-10 production — this is a real immunological axis, validating that ablation extracts genuine signal when handed reasonable candidates.

## 4 Why Triplet Detection Is Hard: The Current Open Challenge

The session’s central diagnostic is that *the cellular-relay hypothesis is testable only after candidate triples  $(A, B, T)$  are proposed*. Three pair-selection strategies were tried:

Table 4: Pair-selection strategy comparison. Recall@5% on KNOWN\_CASCADES; “forward” =  $(A \rightarrow B)$  ordering preserved; “any” = either direction matches.

| Strategy                  | Stage 3 max-W<br>global top-5% | Stage 3 mean-W<br>global top-5% | Stage 3 mean-W<br>per-source top-3 |
|---------------------------|--------------------------------|---------------------------------|------------------------------------|
| Forward recall            | 4/11                           | 4/11                            | 0/11                               |
| Any-direction recall      | 4/11                           | 4/11                            | 2/11                               |
| Pairs to ablate (per exp) | 400                            | 400                             | 270                                |

### 4.1 Why the geo method drops known cascades from its top

The geo bias score measures *how much* cells of type  $T$  shift in the direction of cytokine  $B$  in an  $A$ -treated tube. Two known properties of the dataset distort this score:

1. A few cytokines (IL-32- $\beta$  above all) produce *enormous* cell-type-specific shifts in cell-type embeddings. These dominate every source’s top-K simply because they are geometric outliers.
2. Known cascade partners (e.g., IL-12 / IFN- $\gamma$ , IL-1 $\beta$  / IL-6) involve real but subtler shifts. Their pair score is real but middle-of-the-pack, so they get displaced by the outliers.

**Symmetry asymmetry.** Even when a known cascade has any geometric signal, the geo direction is rarely the canonical one. For instance, the dominant direction in the data is “IFN- $\gamma \rightarrow$  IL-12” (because IFN- $\gamma$ -treated cells shift toward where IL-12 sits), not “IL-12  $\rightarrow$  IFN- $\gamma$ .” This is a property of *latent space geometry*, not biology: the cytokine that produces the bigger response also appears as the geometric “source.” Direction therefore should not be inferred from geo *at all* — this is the ablation method’s job (Section 2.2).

### 4.2 Architecture of the proposed alignment method

The inner-product / cosine method targets the pair-selection bottleneck directly:

- No directionality assumption (symmetric score).
- Two metrics tested: cosine (angular) and raw inner product (magnitude-weighted).
- Three dimensionalities tested: 128D (full), 24D (mild PCA), 6D (aggressive PCA — testing the hypothesis that most dimensions are noise).
- Output is fully compatible with the existing ablation pipeline: hand the top alignment pairs to ablation for directionality.

#### Submission status at session end.

The alignment-based detection code is fully implemented, tested locally (5/5 unit tests pass), committed to `main` (commit 80f54bc), and synced to the cluster. The SLURM submission chain ran into intermittent OTP/auth failures during this session; empirical results comparing the six variants against KNOWN\_CASCADES are pending the next session’s job execution.

## 5 Implementation Summary

### 5.1 Codebase changes from this session

Table 5: Files added or modified.

| File                                                            | Purpose                                                                                                                                                                                                 |
|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>cytokine_mil/analysis/encoder_cache.py</code><br>(new)    | Shared helpers: <code>build_cache</code> , <code>compute_pbs_centroids</code> , <code>_embed_tube</code> , <code>_load_label_encoder</code> . Extracted from <code>run_geo_extract.py</code> for reuse. |
| <code>cytokine_mil/analysis/pair_alignment.py</code><br>(new)   | Core alignment computation: per-(C,T,d) PBS-RC centroids, PCA projections, pair scoring under cosine and inner-product metrics, top-pairs formatting.                                                   |
| <code>scripts/run_geo_extract.py</code> (refactored)            | Now imports cache helpers from <code>encoder_cache.py</code> .                                                                                                                                          |
| <code>scripts/run_inner_product_pairs.py</code><br>(new)        | Per-experiment alignment runner. Six variant outputs per experiment.                                                                                                                                    |
| <code>scripts/analyze_inner_product_results.py</code><br>(new)  | Cross-experiment recall@K analysis, dim comparison plots.                                                                                                                                               |
| <code>scripts/filter_top_pairs.py</code> (modified)             | Switched ranking from max-W to mean-W across cell types.                                                                                                                                                |
| <code>scripts/analyze_two_stage_results.py</code><br>(modified) | Threshold sweep now uses mean-W (with fallbacks).                                                                                                                                                       |
| <code>slurm/inner_product/01_run.slurm</code><br>(new)          | Array job 0-7, 32 GB, 4 CPUs, 1 h.                                                                                                                                                                      |
| <code>slurm/inner_product/02_analyze.slurm</code><br>(new)      | Cross-experiment analysis, 8 GB, 30 min.                                                                                                                                                                |
| <code>slurm/inner_product/meta_submit.slurm</code><br>(new)     | Chains 01 $\rightarrow$ 02 via <code>sbatch -dependency=afterok</code> .                                                                                                                                |
| <code>tests/test_pair_alignment.py</code> (new)                 | 5 unit tests: shape, symmetry, identity-cosine, PCA dims, schema.                                                                                                                                       |

## 5.2 Reusable utilities

# 6 Conclusions and Forward Plan

## 6.1 What works

- C1. Cell-type ablation is the most reliable component of the pipeline.** Given a viable pair, ablation tests directionality directly via the model’s own predictions. It is computationally expensive but biologically faithful. **Future work should treat ablation as the “ground truth” direction call** and focus on improving upstream pair selection.
- C2.** The Stage-3 MIL-finetuned encoder doubles known-cascade recall over the Stage-1 encoder. Future work should use `model_stage3.pt`.
- C3.** Mean-W across cell types is a far better pair ranking metric than max-W (which saturates) or median-W (which has poor pair-level spread).
- C4.** The pipeline is now properly modularized: `encoder_cache.py` holds the shared cache logic, so new pair-detection methods only need to implement scoring on the centroid dictionary.

## 6.2 What does not work yet

- X1. Pair selection from the geo bias score misses known cascades.** Per-source top-3 yields 0/11 forward recall; global top-5% yields 4/11 in any direction. The latent geometry contains too many strong outlier cytokines (notably IL-32- $\beta$ ) that dominate the rankings.
- X2. The cellular-relay triplet  $(A, B, T)$  is the right biological unit, but our current scores are pair-level and only weakly  $T$ -conditional.** The relay cell type is recovered as an argmax, with no calibration on its plausibility.

Table 6: Key utilities, with function signatures, that future work should reuse rather than reimplement.

| Location                      | Signature                                                                                                                               |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| encoder_cache.py              | compute_pbs_centroids(encoder, manifest, gene_names, device) $\rightarrow$ Dict[cell_type, ndarray]                                     |
| encoder_cache.py              | build_cache(encoder, manifest, gene_names, label_encoder, device) $\rightarrow$ list[dict] (each tube as {H, label, cell_types, donor}) |
| pair_alignment.py             | compute_per_atd_centroids(cache, label_encoder, pbs_ct_means, train_donors) $\rightarrow$ Dict[(C,T,d), ndarray]                        |
| pair_alignment.py             | fit_pca_projections(centroids, n_components_list) $\rightarrow$ Dict[dim, Dict[(C,T,d), ndarray]]                                       |
| pair_alignment.py             | compute_pair_scores(centroids, metrics) $\rightarrow$ (pair_scores, relay_T, full_table)                                                |
| latent_geometry.py            | compute_directional_bias_per_donor(cache, le, pbs_ct, train_donors, direction_mode)                                                     |
| latent_geometry.py            | test_directional_significance(bias_result, le, alpha)                                                                                   |
| analyze_cell_type_ablation.py | Ablation entry point; consumes top_pairs.json schema {A,B,relay_cell_type,...}.                                                         |

**X3.** Direction information from geo is unreliable. Asymmetry between ( $A \rightarrow B$ ) and ( $B \rightarrow A$ ) geo scores reflects which cytokine has the bigger response, not the direction of the cascade.

### 6.3 Specific next actions for the future session

- N1.** Run the alignment-based detection pipeline submitted at session end (slurm/inner\_product/meta\_submit.slurm). Compare the six (metric, dim) variants against KNOWN\_CASCADES via analyze\_inner\_product\_results.slurm  $\rightarrow$  recall\_curves.png.
- N2.** If a variant achieves  $\text{recall@5\%} \geq 4/11$ , hand its top\_pairs\_\*.json to the existing ablation pipeline (scripts/analyze\_cell\_type\_ablation.py) for direction calling.
- N3.** Diagnose IL-32- $\beta$  specifically: investigate whether it is a real biological anchor or a training-set artifact (donor-specific? batch effect? unusual transcriptional signature in the encoder).
- N4.** Consider running ablation on a deliberately constructed pair set containing the 11 known cascades, to establish a clean “ablation recovers direction” benchmark independent of pair selection.

## Appendix: Key Quantitative Snapshots

Table 7: Quantitative summary table for cross-method comparison (Stage 3 encoder, exp\_0\_seed42).

| Method                    | Recall@5%      | Score range     | Discrimination | Direction call          |
|---------------------------|----------------|-----------------|----------------|-------------------------|
| geo max-W                 | 4/11 (any dir) | saturates at 55 | poor           | implicit, unreliable    |
| geo median-W              | 2/11           | 5–50            | moderate       | ditto                   |
| geo mean-W                | 4/11           | 23–38           | good           | ditto                   |
| per-source top-3 (mean-W) | 0/11 (fwd)     | ditto           | –              | dominated by outliers   |
| ablation only             | –              | –               | –              | <b>direct, reliable</b> |
| inner-product (designed)  | TBD            | TBD             | TBD            | <b>none (symmetric)</b> |

**Training donors.** D1, D4, D5, D6, D7, D8, D9, D10, D11, D12 (10 donors).

Table 8: The eight experiments and their seeds.

| <b>Exp index</b> | <b>Seed</b> |
|------------------|-------------|
| 0                | 42          |
| 1                | 123         |
| 2                | 7           |
| 3                | 456         |
| 4                | 789         |
| 5                | 314         |
| 6                | 271         |
| 7                | 618         |

**Validation donors (held out).** D2, D3 (2 donors).